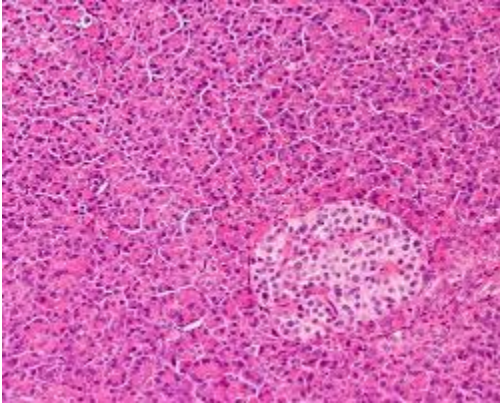


Histology of Endocrine Glands

1. A 34-year old woman comes to the physician for a follow-up examination. She has a history of multiple endocrine neoplasia syndrome type 1. Her biopsy results show new neoplasia (tumors) forming within the tissue shown in the attached light micrograph. Which of the following best represents the gland from which this tissue was taken?



- a. Adrenal
 - b. Pancreas
 - c. Parathyroid
 - d. Pituitary
 - e. Thyroid
2. A scientist is researching the secretions of the endocrine pancreas. The study design requires the scientist to accurately identify the pancreatic endocrine cells with light microscopy techniques. Which of the following methods will be best for detecting the cells of interest?
 - a. Immunohistochemistry
 - b. Reaction with eosin
 - c. Reaction with hematoxylin
 - d. Reactivity with silver stain
 - e. Reactivity with PAS
 3. An 8-year-old boy is brought to the physician by his parents because of a 1-week history of an 8 kg (17lbs) weight loss following a flu-like illness. Physical examination shows a tired looking child with clinical signs of dehydration. A urinalysis shows glucosuria. Fasting blood glucose is 220 mg/dL (normal 50 – 100 mg/dL). Which of the following cells are most likely affected in this case?
 - A. Beta
 - B. Alpha
 - C. Delta

- D. F
- E. PP

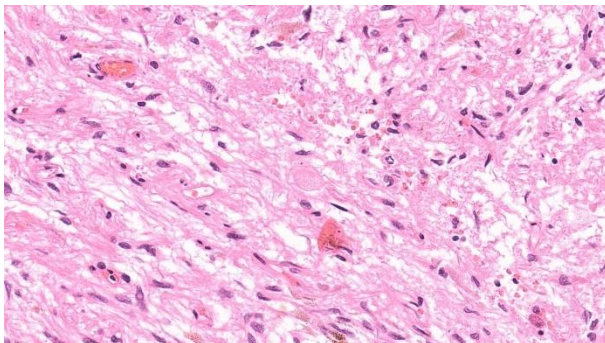
4. A 24-year-old woman comes to the physician for a follow-up examination. 6 weeks ago, she delivered a healthy baby girl. The post-delivery period was complicated by post-partum hemorrhage treated with blood transfusion. Today she complains of extreme tiredness, muscle weakness, failure to breastfeed the newborn, and the absence of menstruation. The physician suspects that there is post-partum injury to an endocrine gland. Which of the following glands is most likely affected?

- A. Parathyroid
- B. Thyroid
- C. Adrenal
- D. Pituitary
- E. Pancreas

5. A 68-year-old man comes to the physician because of a 1-day history of difficulty passing urine and a 2-month history of bone pain. His vital signs are within normal limits. Physical examination shows no abnormalities. Laboratory studies show elevated serum calcium levels. The physician suspects excess of a hormone produced by cells derived from the third and fourth pharyngeal pouches. Which of the following best identifies these cells?

- A. Spongiocytes
- B. Chief cells
- C. Oxyphils
- D. Leydig
- E. Follicular

6. A 50-year-old man is brought to the emergency department following a motor vehicular accident where he sustained severe head injuries. He dies despite appropriate medical care. An autopsy is performed and shows injury to the tissue shown in the attached light micrograph. Which of the following best identifies the location of the injury?



- a. Median eminence
- b. Pars intermedia
- c. Pars nervosa
- d. Pars distalis
- e. Infundibulum

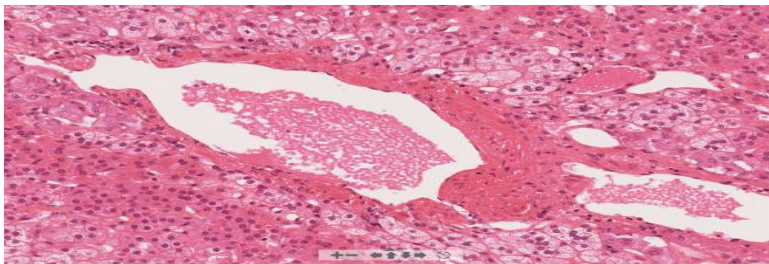
7. A 28-year-old pregnant multigravida woman comes for an obstetrics check-up. She reports that she has been having difficulty with peripheral vision. Clinical imaging is ordered and shows compression of the optic chiasm by an enlarged anterior pituitary gland. Which of the following best describes the embryological origin of the enlarged structure?

- a. Oral cavity
- b. Neuroectoderm
- c. Neural crest cells
- d. Mesoderm
- e. Thyroglossal duct

8. A 29-year-old primigravida woman is admitted to the birthing unit in labor. Over the next two hours, the Obstetrics team becomes concerned with the failure of labor to progress due to inadequate uterine contractions. Which of the following hormones can be administered to enhance uterine contractions?

- a. Estrogen
- b. Parathyroid
- c. Prolactin
- d. Progesterone
- e. Oxytocin

9. A 40-year-old woman comes to the emergency department because of a 1-month history of episodic headaches and dizziness. Her blood pressure is 156/99 mmHg and pulse is 110/min. A CT scan shows a focal lesion in the left suprarenal gland. Biopsy of the lesion shows a pathology associated with the parenchymal cells surrounding the vessel in the attached light micrograph. Which of the following cells is most likely involved in this disease?



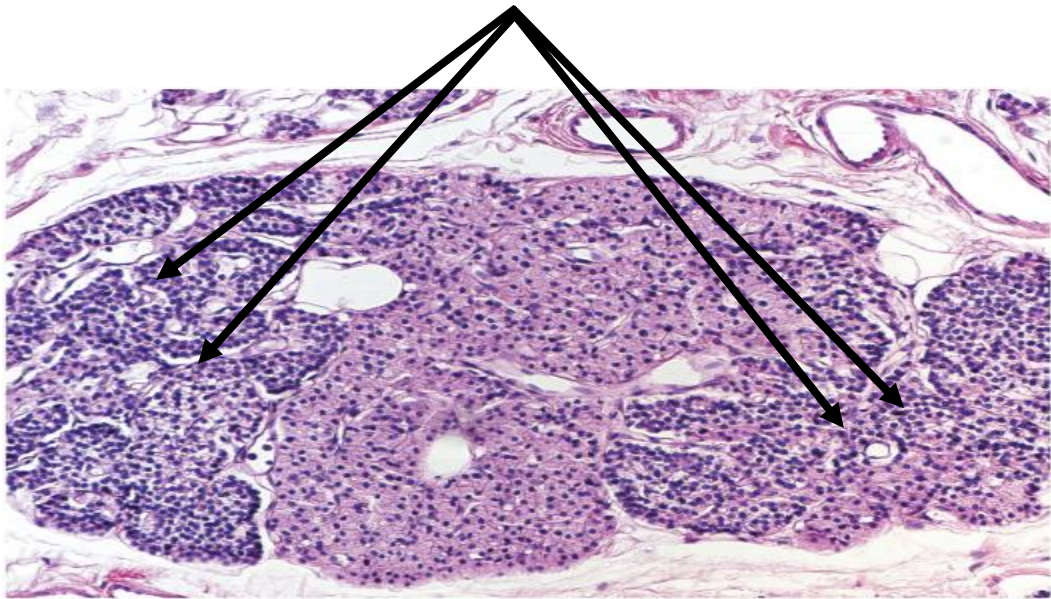
- A. Chromaffin
- B. Chief

- C. Oxyphil
- D. Beta
- E. Spongicytes

10. A 53-year-old man comes to the physician because of a 3-month history of muscle weakness and dizziness. His blood pressure is 190/100 mmHg. Physical examination shows no abnormalities. Laboratory studies show elevated aldosterone and low serum renin levels. An endocrine adenoma is suspected. Which of the following is the most likely location of this pathology?

- a. Anterior pituitary; corticotropes
- b. Adrenal cortex; zona glomerulosa
- c. Adrenal medulla; chromaffin cells
- d. Adrenal cortex; zona fasciculata
- e. Adrenal cortex; zona reticularis

11. A 69-year-old woman comes to the physician because of a 4-month history of generalized weakness, lethargy, weight loss and constipation. She has a history of multiple endocrine neoplasia syndrome diagnosed 6 months ago. She dies despite appropriate medical care. The parathyroid gland is collected for histopathological examination. A light micrograph of the gland is shown in the attached micrograph. Which of the following best describes the cells indicated in at the tip of the arrow?



- a. Chromaffin

- b. Chief
- c. Oxyphil
- d. Parafollicular
- e. Follicular

12. A 40-year-old woman comes to the physician because of a 2-month history of lethargy, cold insensitivity and uncontrolled weight gain. Her blood pressure is 101/59 mmHg and pulse is 58/min. Physical examination shows a soft swelling at the base of the anterior neck. Laboratory studies show decreased T3 and T4 and elevated TSH serum levels. The physician suspects disease of the thyroid gland. Which of the following is most likely associated with the epithelial cells that form the normal structure of this organ?
- a. Microvilli
 - b. Well-developed sER
 - c. Concretions
 - d. Glycogen filled vacuoles

Answers:

1. **The answer is B: Pancreas.** This autopsy specimen shows an islet of Langerhans in the pancreas, surrounded by exocrine acini. The pancreas contains millions of islets, primarily in the tail of the pancreas. These microglands regulate blood glucose levels. The functional cell types in the islets of Langerhans include alpha (α), beta (β), and delta (δ) cells. Each cell type produces a different polypeptide hormone (e.g., insulin, glucagon, or somatostatin). Patients with multiple endocrine neoplasia (MEN) syndromes have gene mutations that make them susceptible to neoplasia or hyperplasia of multiple organs. For example, patients with MEN 2A (Sipple syndrome) develop medullary carcinoma of the thyroid and pheochromocytoma of the adrenal medulla. One-third of these patients also exhibit hyperparathyroidism as a result of parathyroid hyperplasia or parathyroid adenoma. None of the other choices feature the distinctive morphology of pancreatic islets.
2. **The answer is A: Immunostaining property.** Pancreatic islets contain a variable number of cells, ranging from only a few cells, to many hundreds of cells. Each endocrine cell typically secretes one polypeptide hormone. Monospecific antibodies directed to unique polypeptide chains can be used to identify specific hormone-producing cells (e.g., insulin- or glucagon-producing cells). None of the other choices provide a reliable basis for distinguishing between types of functional cells in the endocrine pancreas.
3. **The answer is A: Beta cells.** Type 1 diabetes mellitus (T1DM) is a lifelong disorder of glucose homeostasis that results from the autoimmune destruction of the beta (β) cells in the islets of Langerhans. The clinical onset of T1DM often coincides with another acute illness, such as a febrile viral or bacterial infection.

The disease is characterized by few (if any) functional beta cells in the islets of Langerhans and limited or absent insulin secretion. Alpha cells secrete glucagon. Delta cells secrete somatostatin. F cells (also termed PP cells) secrete pancreatic polypeptide, which stimulates gastric chief cells and inhibits intestinal motility.

4. **The answer is D: Pituitary gland.** This patient experienced postpartum ischemic necrosis of the pituitary gland (Sheehan syndrome). This rare complication of labor and delivery may occur following a severe drop in blood pressure (hypotension) induced by postpartum hemorrhage. The pituitary is particularly susceptible at this time, because its enlargement during pregnancy renders it vulnerable to a reduction in blood flow. The varied clinical manifestations of Sheehan syndrome are related to panhypopituitarism. These clinical manifestations include pallor (decreased MSH), hypothyroidism (decreased TSH), failure of lactation (decreased prolactin), adrenal insufficiency (decreased ACTH), and ovarian failure (decreased FSH and LH). Injury to the other endocrine organs would not lead to the constellation of hormonal insufficiencies observed in this patient.
5. **The answer is B: Chief cells.** Hormones are molecules that “set in motion” the metabolic activities of cells, tissues, and organs. These regulatory molecules (hundreds of unique molecules) are typically released into the circulatory system for distribution throughout the body (endocrine signaling). Hormones may also travel through hydrated interstitial space (tissue compartments). Hormones are classified chemically as steroids, polypeptides, or amino acids. They exhibit target specificity, because nuclear and cell surface receptors display restricted patterns of developmental expression. Neuroendocrine cells, such as parafollicular cells of the thyroid, typically store hormones within membrane bound dense core granules. These secretory granules provide a useful EM marker for the diagnosis of certain poorly differentiated tumors. The other endocrine cells synthesize steroid hormones and do not contain dense core, membrane-bound secretory granules. In this case, the chief cells of the parathyroid gland are secreting PTH, which, through various mechanisms, raise or elevate blood calcium levels. Follicular cells surround the thyroid follicles, and make thyroid hormones.
6. **The answer is C: Pars nervosa of the neurohypophysis.** The pituitary gland (hypophysis) is located at the base of the brain within the sella turcica (L: Turkish saddle). This “master gland” has two components: the posterior lobe (neurohypophysis) and the anterior lobe (adenohypophysis). The posterior lobe of the pituitary shown in the image is composed of lipid-rich (pale staining) nonmyelinated axons, supporting glial cells, and capillary endothelial cells. The hypothalamus is not visible in the image but is located superior to the pituitary in the brain. Hormonal secretions of the hypothalamus regulate pituitary function. The anterior lobe of the pituitary features small cords of glandular epithelial cells. None of the other choices exhibit the distinctive morphologic features of the neurohypophysis.

7. **The answer is A: Oral (pharyngeal) ectoderm/oral cavity.** The anterior pituitary is derived from an evagination of the oropharynx during embryonic development that is referred to as Rathke pouch. It is named after Martin Rathke, who first described it in 1839. Rathke pouch makes contact with a downgrowth from the diencephalon to form the pituitary gland. Remnants of Rathke pouch are preserved as cystic epithelial structures located between the pars nervosa and the pars distalis. These cystic structures can give rise to epithelial tumors that invade and compress adjacent tissues. Destruction of the posterior lobe of the pituitary, the source of ADH, causes diabetes insipidus. One-fourth of the cases of diabetes insipidus are associated with brain tumors, particularly craniopharyngiomas. This tumor arises above the sella turcica from remnants of Rathke pouch. None of the other choices contribute to the formation of the pituitary gland during embryonic development.
8. **The answer is E: Oxytocin.** This hormone is produced by the magnocellular neurons (specifically the paraventricular nucleus) of the hypothalamus. It acts on the smooth muscle cells of the uterus, promoting expulsion of the fetus during labor and delivery. It also acts on the mammary glands to aid in ejection of the milk during breast feeding.
9. **The answer is A: Chromaffin cells.** The adrenal (suprarenal) glands are small, flat organs, embedded in fat near the superior poles of the kidneys. This photomicrograph of the adrenal gland shows a pink-stained cortical region surrounding a pale-stained central (medullary) region. The parenchymal cells of the adrenal medulla are termed chromaffin cells. These large epithelioid cells secrete catecholamines—commonly referred to as the “flight-or-fight” hormones. The adrenal medulla has a dual blood supply that arises from (1) medullary arterioles that bypass the adrenal cortex and (2) capillary sinusoids that perfuse the adrenal cortex (adrenocortical sinusoids). The central adrenomedullary vein drains to the inferior vena cava. None of the other choices describe the parenchymal cell of the adrenal medulla.
10. **The answer is B: Adrenal cortex, zona glomerulosa.** Malignant hypertension and elevated serum levels of aldosterone suggest a diagnosis of Conn syndrome. Primary hyperaldosteronism (Conn syndrome) reflects inappropriate secretion of aldosterone by an adrenal adenoma (75% of cases) or hyperplastic adrenal glands (25% of cases). Most patients with primary aldosteronism are diagnosed after the detection of asymptomatic hypertension. Muscle weakness and fatigue are due to the effects of potassium depletion on skeletal muscle physiology. None of the other tissues secrete aldosterone.
11. **The answer is B: Chief cells.** The large eosinophilic cells in the image are oxyphil cells. These epithelial cells represent a minor cell population in parathyroid glands. They are non secretory cells, and their function (if any) remains to be determined. Chief cells are the principal cells of the parathyroid glands. They are visible in this image as densely packed polygonal cells with

basophilic central nuclei (at the arrows). Follicular and parafollicular cells are found in the thyroid gland.

- 12. The answer is A: Microvilli.** The thyroid gland is composed of numerous hollow, spherical structures (follicles) lined by a simple epithelium. The apical surface of these cells faces the follicular lumen and possess microvilli, whereas the basal surface of these cells rests on an extracellular basal lamina. Follicular epithelial cells respond to TSH, which is secreted by the anterior pituitary. Follicular cells synthesize, store, and secrete thyroid hormones (T3 and T4). These iodinated, tyrosine-based hormones regulate growth and development, as well as general body metabolism and heat production.